

Customer Shopping Behavior Analysis

An end-to-end analytics report — from raw data to actionable insights — using **Python**, **SQL**, and **Power BI**

ANALYTICS REPORT

RETAIL DATA



Executive Summary

This report analyzes **3,900 retail transactions**— cleaning and enriching data in Python, answering 10 business questions in SQL, and surfacing insights in a Power BI dashboard.

3,900

Transactions Analyzed

18→19

Columns Before / After

10

SQL Questions Answered

1

Power BI Dashboard

Dataset at a Glance

Who Bought It

Age, Gender, Location,
Subscription Status

What Was Bought

Item, Category, Purchase
Amount, Season, Size, Color

How They Shop

Discount, Promo Code, Previous Purchases, Frequency, Review Rating,
Shipping Type



Data quality issue: **37 missing values** in Review Rating — addressed during cleaning.

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Data Preparation in Python

Raw data was cleaned and enriched in Python with pandas before loading into PostgreSQL.

01

Load & Inspect

Read CSV, ran `df.info()` and `df.describe()`

03

Standardize & Engineer

snake_case headers; added `age_group` and `purchase_frequency_days`; dropped duplicate `promo_code_used`

02

Fix Missing Ratings

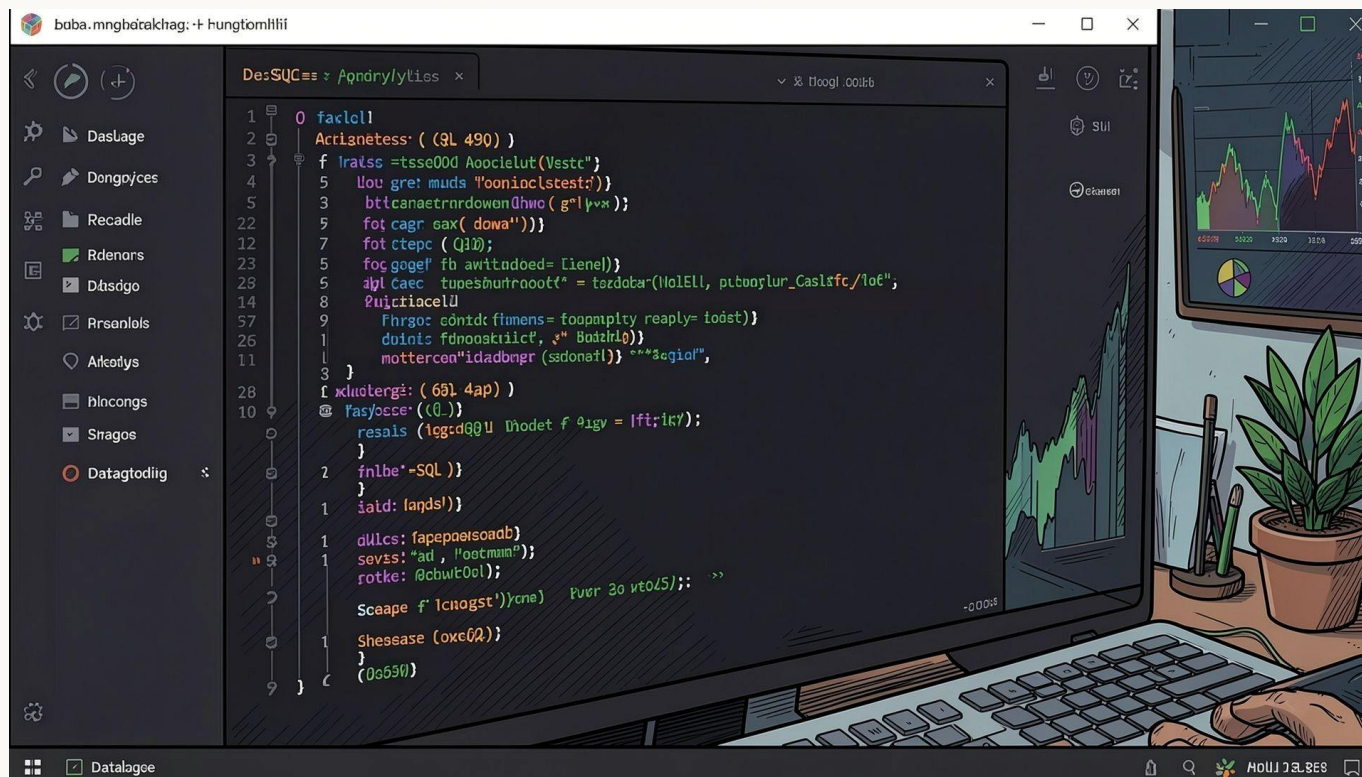
37 nulls filled with **category-level median**

04

Hand Off to SQL

Cleaned DataFrame loaded into PostgreSQL

Business Questions 1–5



Q1: Revenue by Gender

Male: **\$157,890** vs. Female: **\$75,191**

Q2: Discount Users Who Spent Big

839 customers used a discount and spent above average

Q3: Top-Rated Products

Gloves (3.86 ★), Sandals (3.84 ★), Boots (3.82 ★)

Q4: Standard vs. Express Shipping

Express \$60.48 vs. Standard \$58.46
— negligible difference

Q5: Subscribers vs. Non-Subscribers

Avg spend nearly identical (~\$59–60); non-subscribers drive more total revenue by volume

Business Questions 6–10

Q6: Products Most Reliant on Discounts

Hat (50%), Sneakers (49.66%), Coat (49.07%)

Q7: Customer Loyalty Tiers

Loyal: **3,116** · Returning: 701 · New: **83**

Q8: Category Best-Sellers

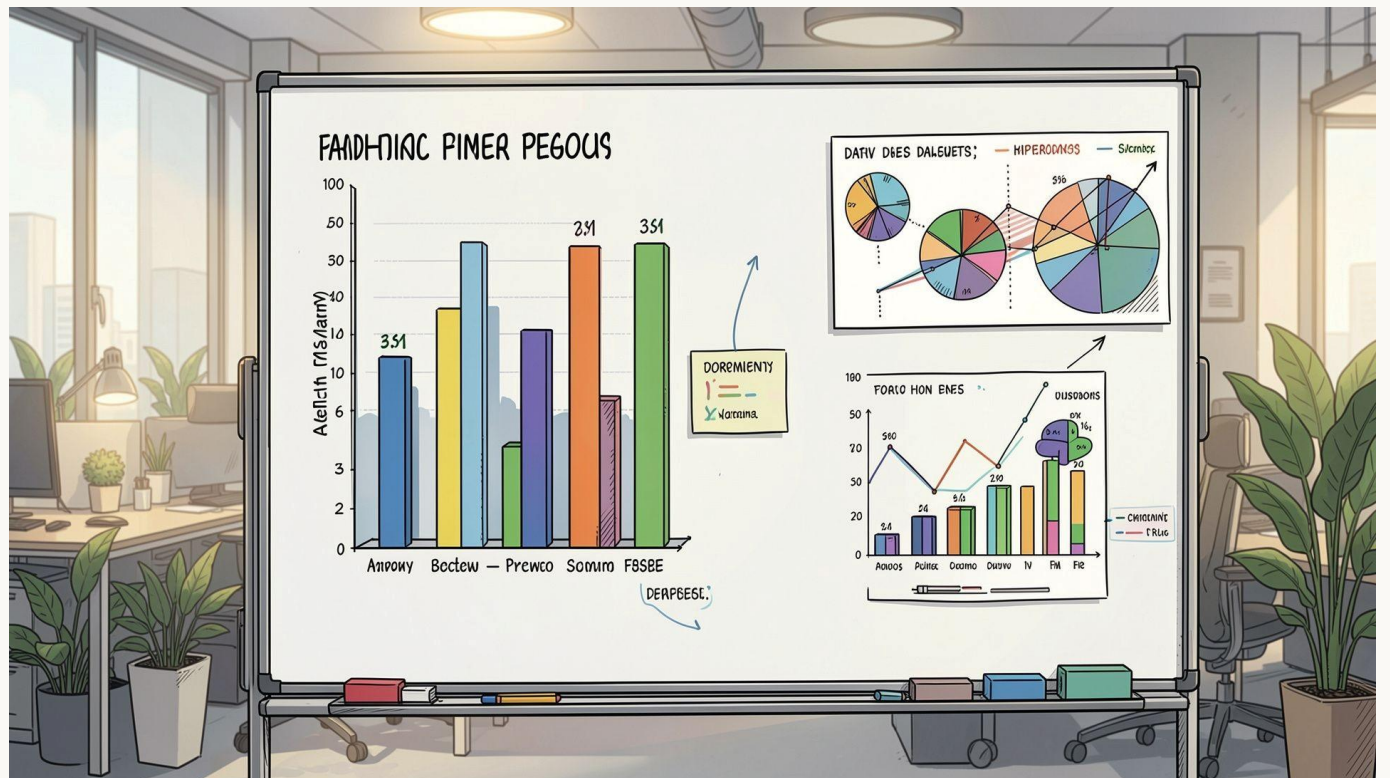
Jewelry, Blouse, Sandals, Jacket lead their categories

Q9: Repeat Buyers & Subscriptions

Repeat buyers (5+ purchases): 2,518 non-subs vs. 958 subs

Q10: Revenue by Age Group

Young Adult: **\$62,143** · Middle-aged: \$59,197 · Adult: \$55,978 · Senior: \$55,763



Power BI Dashboard

A single interactive dashboard provides self-service access to all key metrics — filterable by subscription status, gender, category, and shipping type.

3.9K
Customers
\$59.76
Avg Purchase
3.75★
Avg Review
27%
Subscribers



What the Numbers Say



Gender Revenue Gap

Male revenue is **~2× female** (\$157K vs. \$75K) — warrants category-level follow-up.



Discounts Aren't Just for Bargain-Hunters

839 discount users still spent **above average**.



Retention Is Strong

80% of customers are Loyal (10+ purchases); only 2% are brand-new.



Shipping Speed ≠ Basket Size

Express orders averaged only **~\$2 more** than Standard.

Recommendations

1 Rework the Subscriber Value Proposition

Add clear incentives (free shipping, early access) — subscription isn't lifting spend.

2 Invest in New-Customer Acquisition

98% are repeat/loyal — the funnel needs a dedicated acquisition push.

3 Audit Discount Depth on High-Rate Products

~50% discount rates on Hats and Sneakers may erode margin without driving volume.

4 Lean on Top-Rated Products in Marketing

Feature Gloves, Sandals, and Boots in campaigns; cross-sell with lower-rated items.

5 Investigate the Gender Revenue Gap

The ~\$80K gap warrants a follow-up analysis by category before deciding on campaigns.

Project Stack & Takeaways

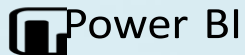
Tools Used



pandas for cleaning &
feature engineering



CTEs,
window functions,
subqueries



Interactive dashboard with filters & KPIs

Key Takeaways

- Male customers drive ~2× the revenue of female customers
- 80% of the base is already Loyal — acquisition is the growth lever
- Subscribers spend the same as non-subscribers — the program needs a stronger value prop
- Top-rated products (Gloves, Sandals, Boots) are underutilized in marketing



Full code, SQL queries, and dashboard available on **GitHub**.